**Page 2: Introduction – What is a Chemical Reaction?**

We live in a world of constant change. Some changes are physical (like melting of ice), and some are chemical. Understanding chemical reactions is fundamental to chemistry.

**What is a Chemical Reaction?**
A **chemical reaction** is a process in which one or more substances (reactants) are transformed into one or more new substances (products) with different properties.

**Examples from Daily Life:**

- Digestion of food in our body.
- Rusting of iron (formation of reddish-brown layer).
- Burning of wood or LPG gas.
- Curdling of milk.
- Photosynthesis in plants.

**Key Concept:**
In a chemical reaction, the atoms themselves are not created or destroyed. They are simply rearranged to form new molecules. This is the basis of the Law of Conservation of Mass.

**Signs of a Chemical Reaction (How to know if one has occurred?):**

1. **Change in State:** Formation of a solid precipitate, or evolution of a gas.
2. **Change in Colour:** E.g., green to black (oxidation), colourless to blue.
3. **Change in Temperature:** The reaction mixture may become hot (exothermic) or cold (endothermic).
4. **Evolution of Gas:** Bubbles may form (e.g., $CO_2$, $H_2$).
5. **Formation of a Precipitate:** An insoluble solid forms in a liquid solution.

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**Page 3: Writing a Chemical Reaction – Word Equations**

Before we use symbols, we first describe the reaction in words.

**Word Equation:**
A word equation represents a chemical reaction using the names of the reactants and products.

**Format:**
**Reactants → Products**

- **Reactants:** Substances that take part in the reaction (written on the left side).

- **Products:** New substances formed as a result of the reaction (written on the right side).
- The arrow (→) means "reacts to form" or "yields".

**Example 1: Burning of Magnesium**

- **Observation:** A magnesium ribbon burns with a dazzling white light, producing a white powder (magnesium oxide).
- **Word Equation:**
  **Magnesium + Oxygen → Magnesium Oxide**
  *(Reactants) (Product)*

**Example 2: Reaction of Zinc with Hydrochloric Acid**

- **Observation:** Zinc granules react with acid, producing bubbles of hydrogen gas and a soluble salt (zinc chloride).
- **Word Equation:**
  **Zinc + Hydrochloric Acid → Zinc Chloride + Hydrogen**
  *(Reactants) (Products)*

Word equations are a good starting point, but they are long and not universal. To communicate precisely, chemists use **chemical equations**.

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**Page 4: Writing a Chemical Equation – Using Symbols**

A **chemical equation** is a shorthand representation of a chemical reaction using chemical formulas and symbols.

**Format:**
**Reactants (formulas) → Products (formulas)**

**Steps to write a skeletal chemical equation:**

1. Write the word equation.
2. Replace the names of the substances with their correct chemical formulas.
3. Separate the reactants and products with an arrow (→).

**Example: Burning of Magnesium**

- Word Equation: Magnesium + Oxygen → Magnesium Oxide
- Formulas: $Mg + O_2 \rightarrow MgO$
- **Skeletal Equation: $Mg + O_2 \rightarrow MgO$**

**Example: Reaction of Zinc with Hydrochloric Acid**

- Word Equation: Zinc + Hydrochloric Acid → Zinc Chloride + Hydrogen
- Formulas: $Zn + HCl \rightarrow ZnCl_2 + H_2$
- **Skeletal Equation: $Zn + HCl \rightarrow ZnCl_2 + H_2$**

**The Problem with Skeletal Equations:**
Look at the equation: **$Mg + O_2 \rightarrow MgO$**

- Left side: 1 Mg atom, 2 O atoms.
- Right side: 1 Mg atom, 1 O atom.
  The number of oxygen atoms is not the same on both sides. This violates the **Law of Conservation of Mass** (mass can neither be created nor destroyed). Therefore, a skeletal equation must be **balanced**.

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**Page 5: Balancing a Chemical Equation – The Concept**

Balancing a chemical equation means making sure the number of atoms of each element is the same on both the reactant and product sides. We do this by adding **coefficients** (numbers written before the formulas).

**The Rules (Conceptual):**

1. **Never change the subscripts** in a chemical formula to balance an equation. Changing the subscript changes the substance itself (e.g., $H_2O$ is water, $H_2O_2$ is hydrogen peroxide – a different chemical).
2. **Only change the coefficients** (the numbers placed in front of the formulas). A coefficient multiplies the entire compound.
3. Balance the equation by trial and error, starting with the element that appears in the fewest compounds.

**Let's balance $Mg + O_2 \rightarrow MgO$:**

| Element | Reactants (Left) | Products (Right) | Balance |
|---|---|---|---|
| **Mg** | 1 | 1 | Balanced |
| **O** | **2** (in $O_2$) | **1** (in MgO) | **Not Balanced** |

- To balance Oxygen, we need 2 O atoms on the right. So, put a coefficient '2' before MgO.

- **$Mg + O_2 \rightarrow 2MgO$**

- Now recount: O is balanced (2 on each side). But Mg is now: Left = 1, Right = 2.
- To balance Mg, put a coefficient '2' before Mg on the left.
  - **$2Mg + O_2 \rightarrow 2MgO$**

Now check:

- Left: 2 Mg, 2 O
- Right: 2 Mg, 2 O

**The equation is now balanced.**

**Final Balanced Equation: $2Mg + O_2 \rightarrow 2MgO$**

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**Page 6: Types of Chemical Reactions – An Overview**

Chemical reactions can be classified into several types based on how the atoms are rearranged. For Class 10, the main types are:

1. **Combination Reaction**
2. **Decomposition Reaction**
3. **Displacement Reaction**
4. **Double Displacement Reaction**
5. **Oxidation and Reduction (Redox Reaction)**

Understanding the *type* of reaction helps us predict the products.

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**Page 7: Combination and Decomposition Reactions**

**1. Combination Reaction**
A reaction in which two or more reactants combine to form a single product.

**General Format: $A + B \rightarrow C$**

- **Example 1 (Burning of Coal):**
  - $C (s) + O_2 (g) \rightarrow CO_2 (g)$
  - Carbon + Oxygen → Carbon Dioxide
- **Example 2 (Formation of Water):**
  - $2H_2 (g) + O_2 (g) \rightarrow 2H_2O (l)$

    - Hydrogen + Oxygen → Water

- **Example 3 (Quicklime to Slaked Lime):**
    - $CaO (s) + H_2O (l) \rightarrow Ca(OH)_2 (aq) + Heat$
    - Calcium oxide (quicklime) reacts with water to produce slaked lime. This is an **exothermic** reaction (releases a lot of heat).

**2. Decomposition Reaction**
A reaction in which a single compound breaks down into two or more simpler substances. It is essentially the opposite of a combination reaction.

**General Format: C → A + B**

Decomposition reactions usually require energy in the form of heat, light, or electricity.

- **Thermal Decomposition (Heat):**
    - **Example:** When ferrous sulphate crystals (green) are heated, they decompose to ferric oxide (brown), sulphur dioxide, and sulphur trioxide.
    - $2FeSO_4 (s) \rightarrow Fe_2O_3 (s) + SO_2 (g) + SO_3 (g)$
- **Electrolytic Decomposition (Electricity):**
    - **Example:** Passing electricity through water breaks it into hydrogen and oxygen.
    - $2H_2O (l) \rightarrow 2H_2 (g) + O_2 (g)$
- **Photochemical Decomposition (Light):**
    - **Example:** The reaction used in photography. Silver chloride turns grey in sunlight, decomposing into silver and chlorine.
    - $2AgCl (s) \rightarrow 2Ag (s) + Cl_2 (g)$
    - $2AgBr (s) \rightarrow 2Ag (s) + Br_2 (g)$ *(used in photography)*

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**Page 8: Displacement and Double Displacement Reactions**

**3. Displacement Reaction**
A reaction in which a more reactive element displaces a less reactive element from its compound.

**General Format: A + BC → AC + B**

- **Example 1 (Iron with Copper Sulphate):**

- **Word Equation:** Iron + Copper Sulphate → Iron Sulphate + Copper
- **Chemical Equation:** Fe (s) + $CuSO_4$ (aq) → $FeSO_4$ (aq) + Cu (s)
- **Observation:** The blue colour of copper sulphate solution fades, and a reddish-brown coating of copper is deposited on the iron nail. This happens because Iron (Fe) is more reactive than Copper (Cu).

- **Example 2 (Zinc with Copper Sulphate):**
  - Zn (s) + $CuSO_4$ (aq) → $ZnSO_4$ (aq) + Cu (s)
  - (Zinc is more reactive than copper).

**4. Double Displacement Reaction**
A reaction in which two different compounds exchange their ions to form two new compounds. This often results in the formation of a precipitate (insoluble solid).

**General Format: AB + CD → AD + CB**

- **Example (Barium Chloride with Sodium Sulphate):**
  - **Word Equation:** Barium Chloride + Sodium Sulphate → Barium Sulphate + Sodium Chloride
  - **Chemical Equation:** $BaCl_2$ (aq) + $Na_2SO_4$ (aq) → $BaSO_4$ (s) + 2NaCl (aq)
  - **Observation:** A white, insoluble solid (precipitate) of Barium Sulphate is formed immediately.

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**Page 9: Oxidation and Reduction (Redox Reactions)**

Many important reactions involve the gain or loss of oxygen or hydrogen.

**Oxidation:**

- **Definition 1:** The gain of oxygen by a substance.
- **Definition 2:** The loss of hydrogen by a substance.
- A substance that gains oxygen (or loses hydrogen) is said to be **oxidized**.

**Reduction:**

- **Definition 1:** The loss of oxygen by a substance.
- **Definition 2:** The gain of hydrogen by a substance.
- A substance that loses oxygen (or gains hydrogen) is said to be **reduced**.

**Redox Reaction:**
Oxidation and reduction always occur **together**. If one substance is oxidized, another must be reduced. Such reactions are called **Redox Reactions** (Reduction-Oxidation).

**Example: Reaction of Copper Oxide with Hydrogen**

- **Equation:** $CuO (s) + H_2 (g) \rightarrow Cu (s) + H_2O (l)$

| Substance | What Happens | Term |
|---|---|---|
| **Copper Oxide (CuO)** | Loses Oxygen (becomes Cu) | **Reduced** |
| **Hydrogen ($H_2$)** | Gains Oxygen (becomes $H_2O$) | **Oxidized** |

Here, CuO is the **oxidizing agent** (it gives oxygen to $H_2$) and $H_2$ is the **reducing agent** (it takes oxygen from CuO).

**Corrosion and Rancidity – Effects of Oxidation:**

- **Corrosion:** The slow destruction of metals due to reaction with moisture, air, and acids.
  - *Example:* Rusting of iron (formation of $Fe_2O_3 \cdot xH_2O$), tarnishing of silver (black $Ag_2S$).
- **Rancidity:** The spoilage of food containing oils and fats due to oxidation, leading to a bad smell and taste.
  - *Prevention:* Adding antioxidants (like BHA, BHT), packing in airtight containers, or flushing with nitrogen gas (to remove oxygen).

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**Page 10: Summary and Revision Points**

| Concept | Key Point |
|---|---|
| **Chemical Reaction** | Transformation of reactants into products with new properties. |
| **Signs of Reaction** | Change in state, colour, temperature; evolution of gas; formation of precipitate. |

| Concept | Key Point |
| --- | --- |
| **Chemical Equation** | Representation using symbols and formulas. |
| **Balanced Equation** | Equal number of atoms of each element on both sides (Law of Conservation of Mass). |
| **Combination Reaction** | Two or more reactants form one product. (A + B → C) |
| **Decomposition Reaction** | One reactant breaks down into two or more products. (C → A + B) Requires energy. |
| **Displacement Reaction** | A more reactive element displaces a less reactive one. (A + BC → AC + B) |
| **Double Displacement** | Ions are exchanged between two compounds. (AB + CD → AD + CB) Often forms a precipitate. |
| **Oxidation** | Gain of Oxygen / Loss of Hydrogen. |
| **Reduction** | Loss of Oxygen / Gain of Hydrogen. |
| **Redox Reaction** | Oxidation and Reduction occur simultaneously. |
| **Corrosion** | Deterioration of metals due to oxidation (e.g., rusting). |
| **Rancidity** | Oxidation of fats and oils in food, causing spoilage. |

**Important Mnemonic:**

- **OIL RIG: O**xidation **I**s **L**oss (of electrons/hydrogen), **R**eduction **I**s **G**ain (of electrons/hydrogen). (For higher classes, but a good concept to remember).

**Final Tip for Board Exams:**
Always write the **states** of reactants and products (s, l, g, aq) in your final balanced equation. This shows a complete understanding of the reaction conditions. Also, when explaining a reaction, always mention the **type of reaction** and any **observable changes** (colour, gas, precipitate).